X-ray report generation using retrieval method and large language models (LLMs)

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Background and motivation

Radiologists are overloaded

Existing report generation methods has limitations

clinically inaccurate(hallucinations)

linguistically influent

Objective and Innovation

Objective: accurate and fluent reports generation

Innovation: **dual-layer** report generation

Retrieve approach for accuracy

LLM refinement for fluency

Automated X-ray report system: method overview

Model approach

- CLIP-based image-text joint modeling
- Goal: directly generates reports from X-ray images

2 × 3 Experimental setup

- **2 models**: Fine-tuned CLIP, BiomedCLIP
- **3 retrieval levels**: images, reports, sentences
- Goal: Identify the most accurate report generation method

System outcome

- End-to-end system
- Inputs X-ray images → outputs clinically usable reports

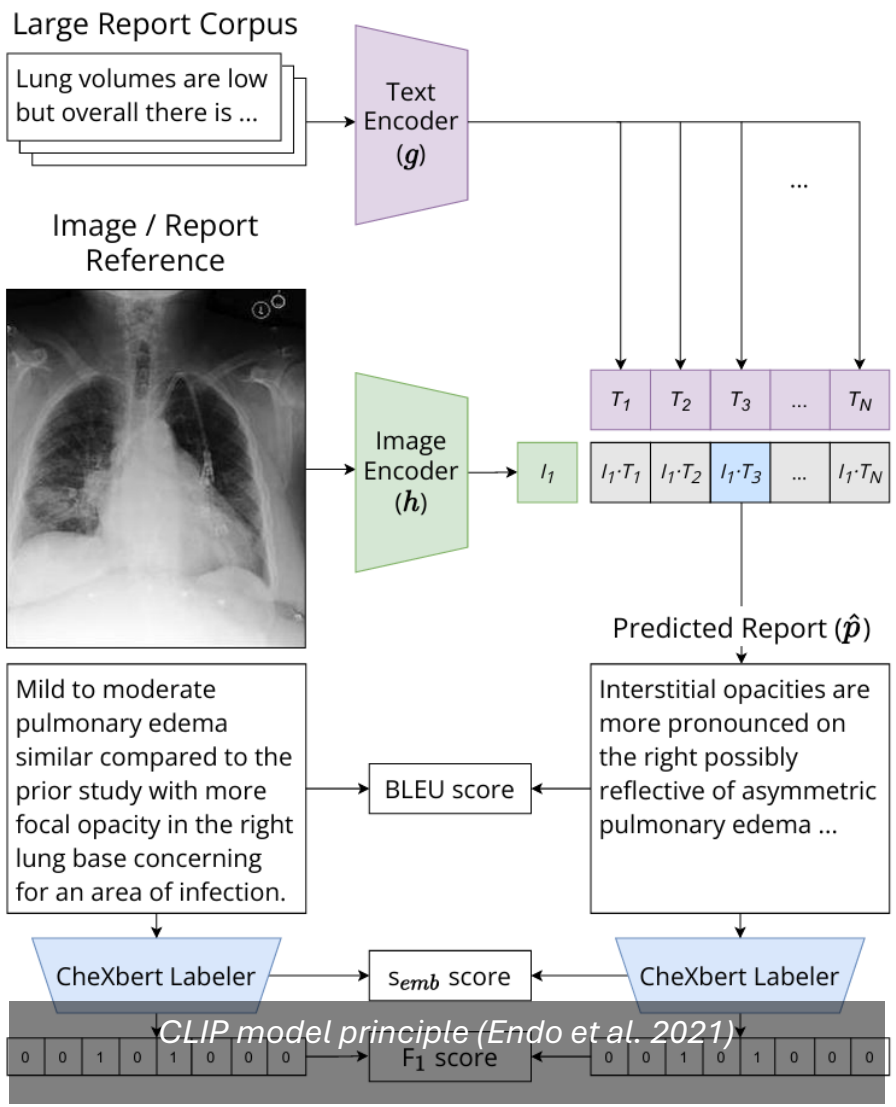
Presentation flow

1. CLIP model principles
2. Algorithm for three retrieval levels
3. System video as demonstration
4. Show experimental results

CLIP model principle

- Text encoder → text embeddings.
- Image encoder → image embeddings.
- Encode image and text into **one shared embedding space for image-text comparison**
- Compare embeddings (cosine similarity)
- Retrieve top-k most similar texts as predicted report.

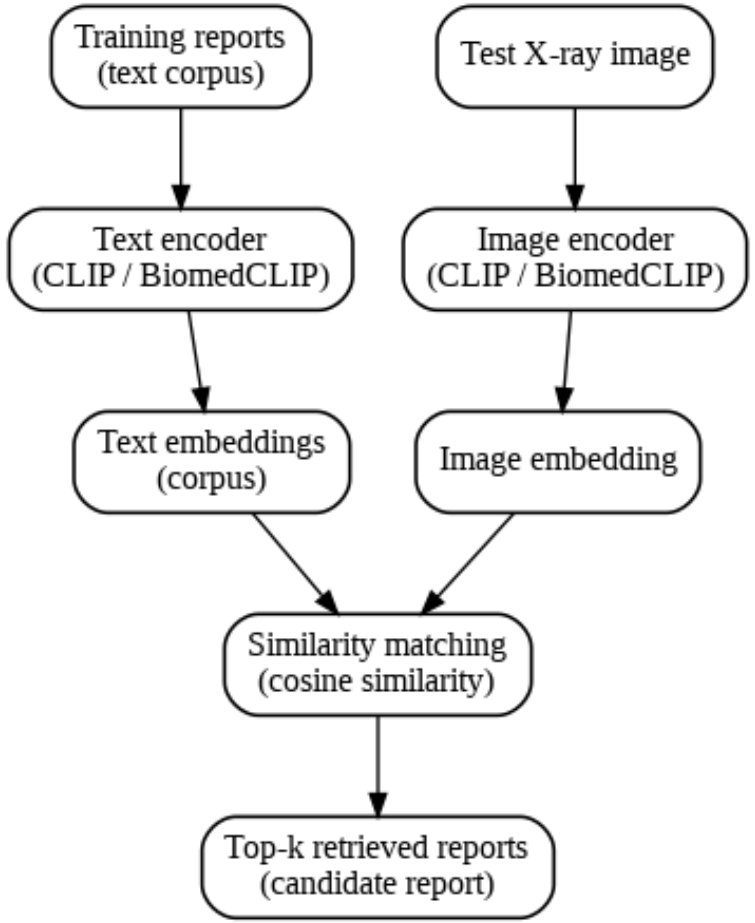
- *Input adaptation: the single channel gray-scale X-ray images are replicated into 3 channels to match the RGB input format of the vision encoder.*



CLIP model principle (Endo et al. 2021)

Retrieval Module (CLIP and BiomedCLIP)

- **Dataset:** IU-Xray (train set → corpus,
- test set → input image)
- **Models:** CLIP & BioMedCLIP for comparison



Retrieval-based report generation pipeline

Retrieval: text-level (reports and sentences)

Text corpus:

- **Report-level:** reports from train set
- **Sentence-Level:** independent sentences, separated from full reports

Process:

- Encode test images into embeddings
- Encode train reports or sentences into embeddings
- Compute cosine similarity
- Retrieve Top-K similar reports or sentences
- Concatenate their reports → generated report

Algorithm 2 CLIP-based Chest X-ray Image to Text Retrieval

Require: X-ray image files $\{I_1, ..., I_m\}$, Text corpus C, Pretrained CLIP model M

Ensure: Top-k relevant texts for each image

```
Initialize CLIP model M
Encode text corpus C into normalized embeddings {V_T}
for each image I_i do
    Encode I_i into normalized embedding V_{I_i}
    Compute similarity scores S_i with {V_T}
    Retrieve top-k texts from corpus C for I_i
end for
return All top-k texts
```

Updated sentence-level (top1 to topk fallback):

- If Top-1 sentence indicates no abnormality → use it as the report
- Else → retrieve Top-5 similar sentences and concatenate as the report

Algorithm 3 Top-1 Sentence Retrieval with CheXbert Fallback

Require: X-ray image embeddings $\{V_{I_i}\}$, Text sentence embeddings $\{V_T\}$, Text sentences $C = \{T_1, ..., T_n\}$, CheXbert model M, fallback k

Ensure: Generated report sentences for each image

```
for each image I_i do
    Compute similarity scores S_i = V_{I_i} · V_T^⊤
    Get top-k sentence indices: topk_i ← topk(S_i, k)
    top1_sentence ← C[topk_i[0]]
    if CheXbert(top1_sentence) indicates No Finding then
        report ← [top1_sentence]
    else
        report ← [C[j] for j in topk_i]          ▷ fallback to top-k sentences
    end if
    Store report for I_i
end for
return Report for images I_i
```

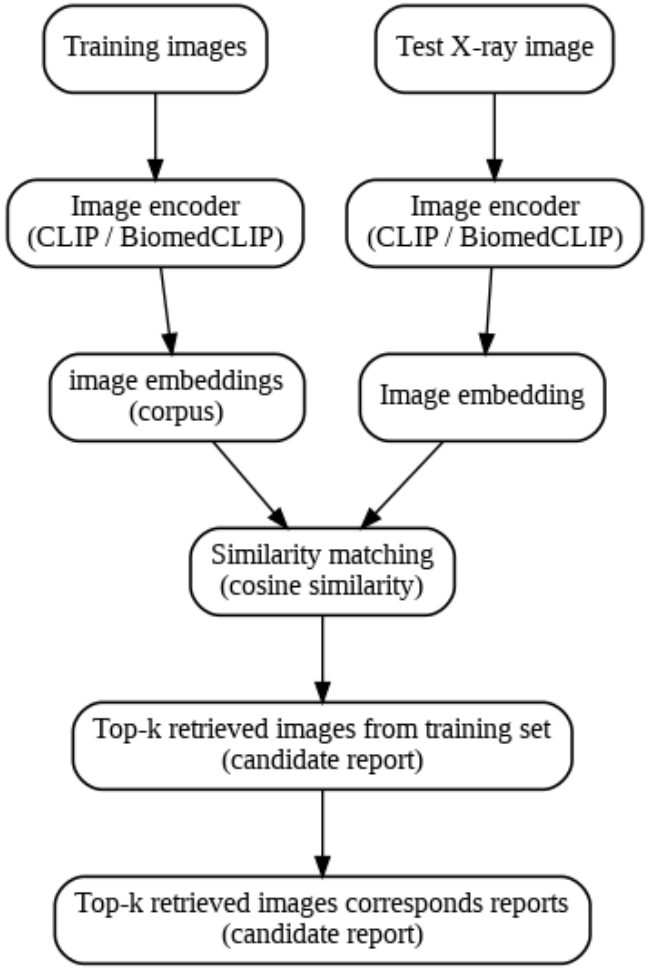
Retrieval: image-level

Algorithm 1 CLIP-based Chest X-ray Image to Image Retrieval

Require: X-ray image files $\{I_1, ..., I_m\}$, images set X, Pretrained CLIP model M

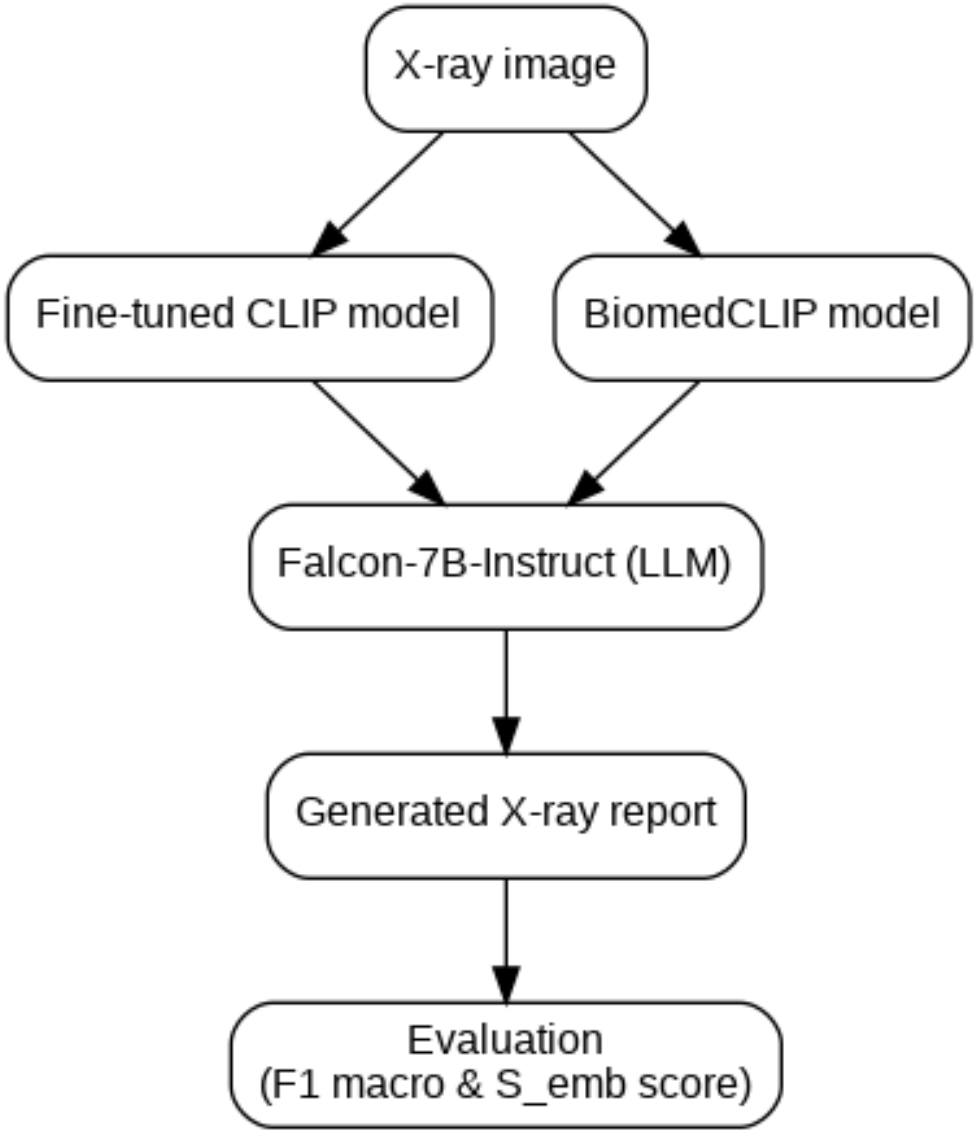
Ensure: Top-k relevant training set images for each test image

```
Initialize CLIP model M
Encode images set X into normalized embeddings {V_T}
for each image I_i do
    Encode I_i into normalized embedding V_{I_i}
    Compute similarity scores S_i with {V_T}
    Retrieve top-k training images from images set X for I_i
end for
return All top-k image
```



System workflow(end-to-end report generation)

- **Input:** Chest X-ray image
- **Retrieval:** Find top-k relevant report texts using CLIP-based embeddings
- **LLM refinement:** Falcon-7B-Instruct improves accuracy and fluency
- **Output:** Final radiology report
- **Evaluation:** F1-macro & S_emb for quality assessment



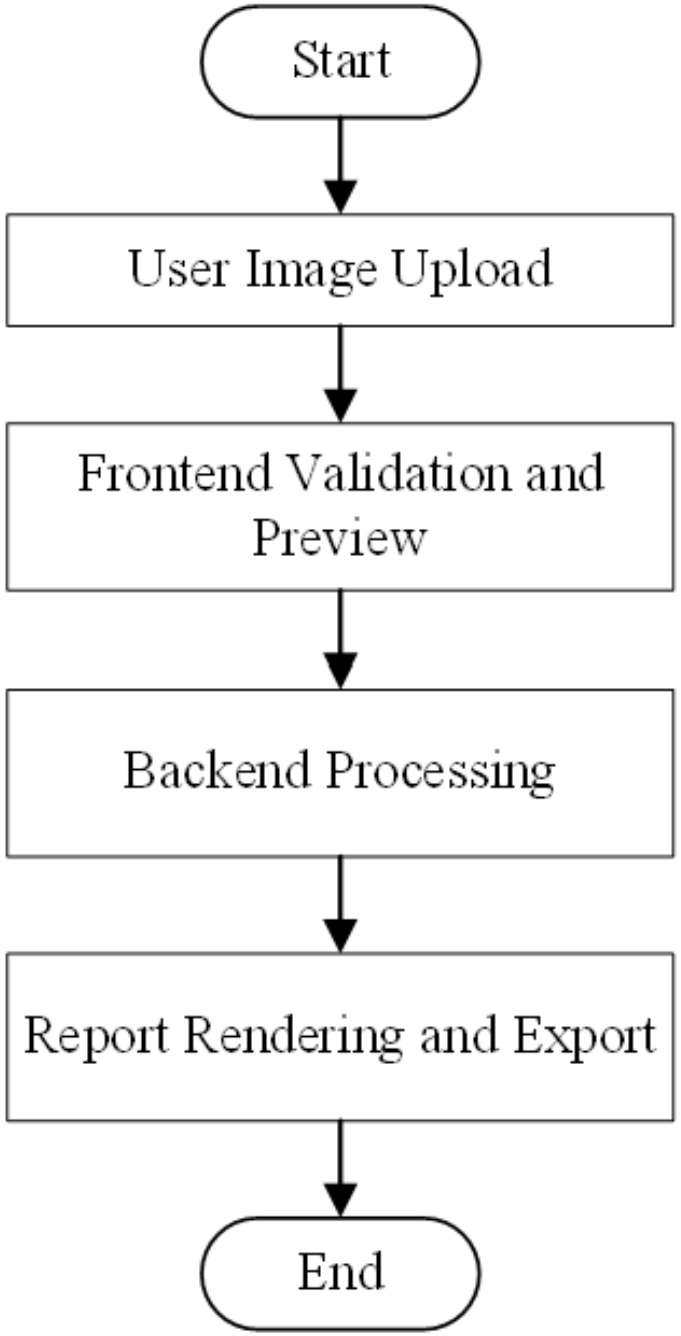
Report generation and evaluation process

System architecture

Upload X-ray image via the frontend

Backend processes the image

generates a report Generated report is displayed back on the frontend



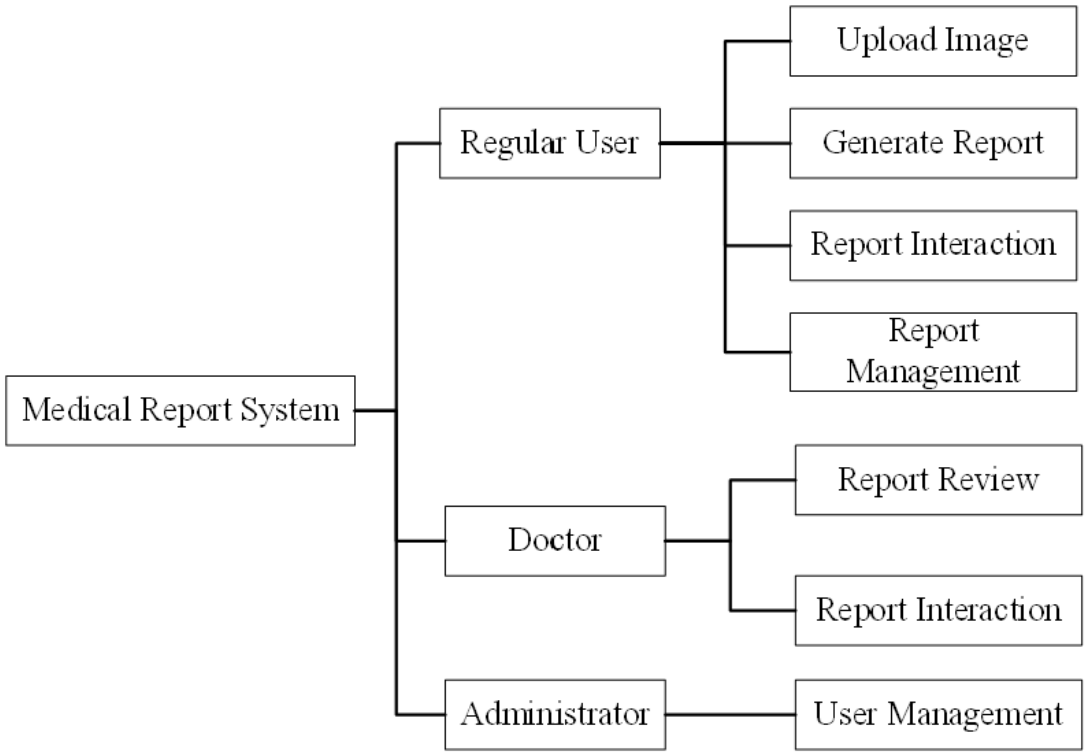
Frontend detailed functions

Roles: doctors, regular user, admin

Regular user(patients) also can use our system to generate X-ray reports

Doctor can review, comment and revise the generated reports.

Admin: manage users and doctors accounts.

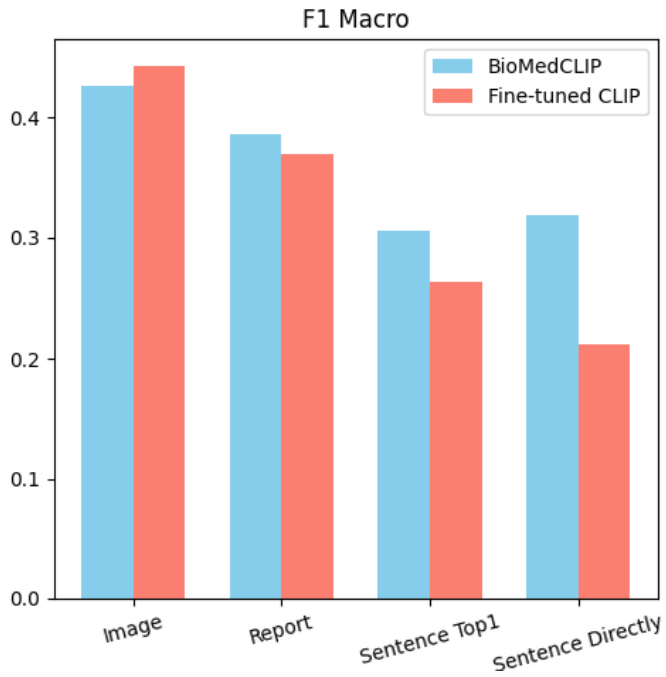


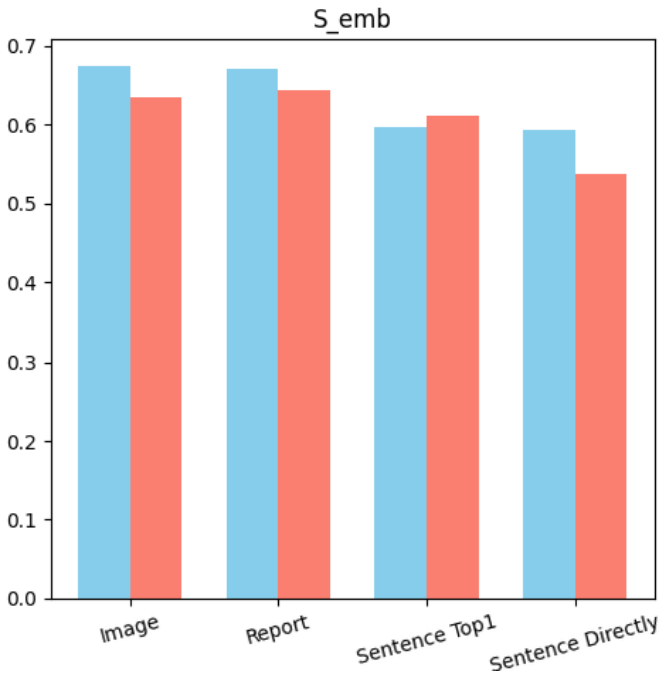
Report evaluation

Clinical accuracy: F1-macro (CheXbert-labeled)

Semantic similarity: S_emb (embedding cosine similarity)

Compare generated reports with ground truth reports



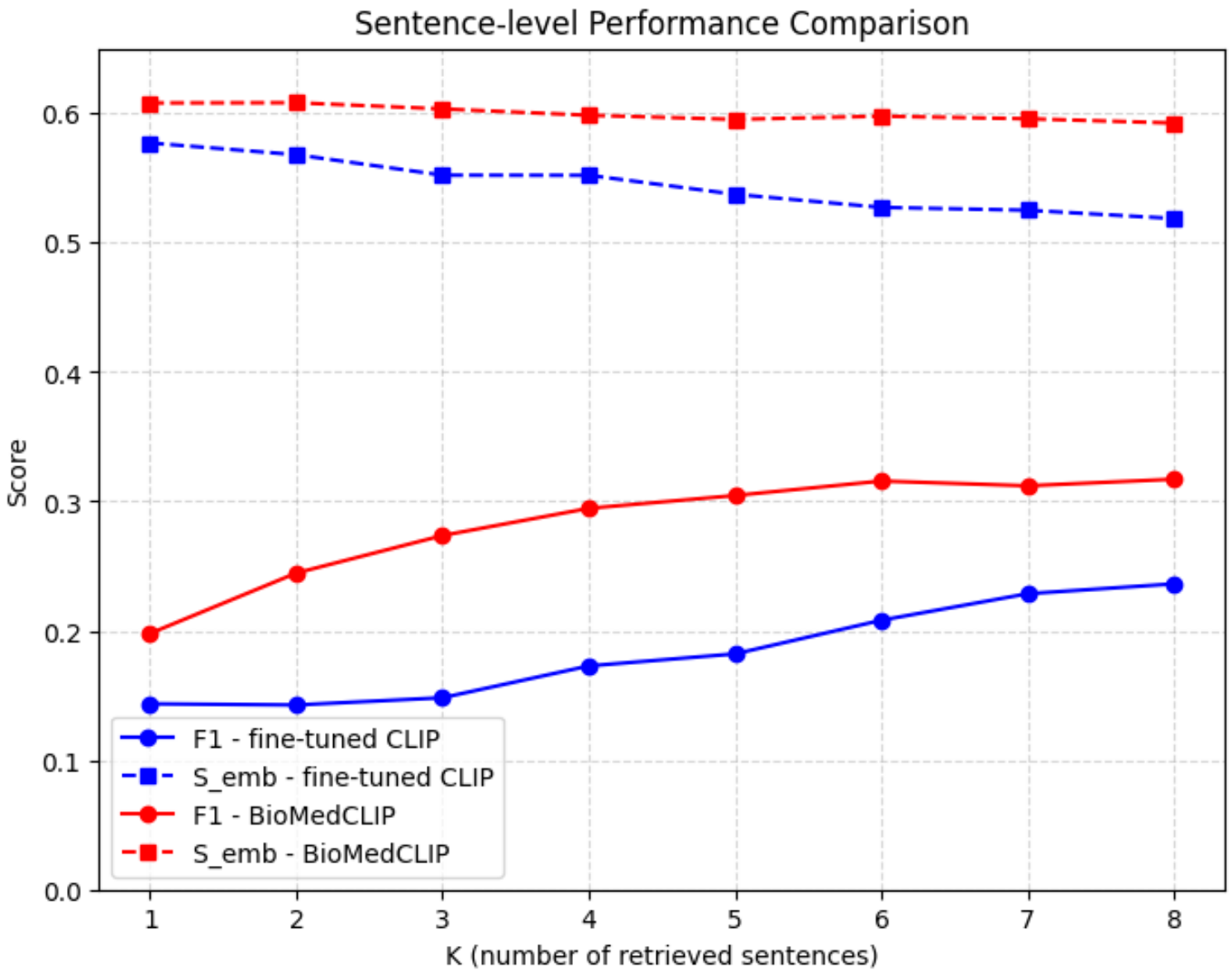


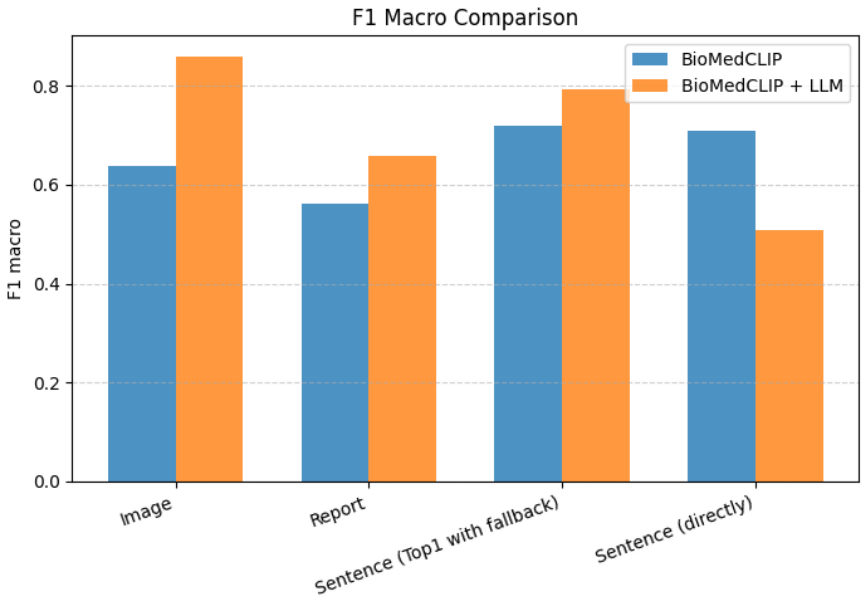
- **BiomedCLIP outperformed the fine-tuned CLIP on both F1- macro and S_emb score**

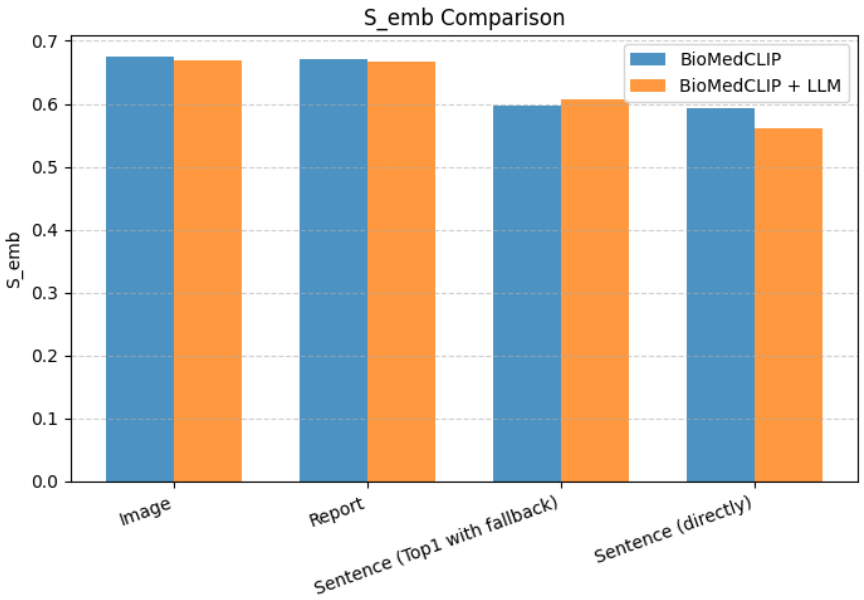
Experiment result (two CLIP model)

BioMedCLIP outperforms the fine-tuned CLIP at every Top-K sentence retrieval setting.

Experiment result (two CLIP model)







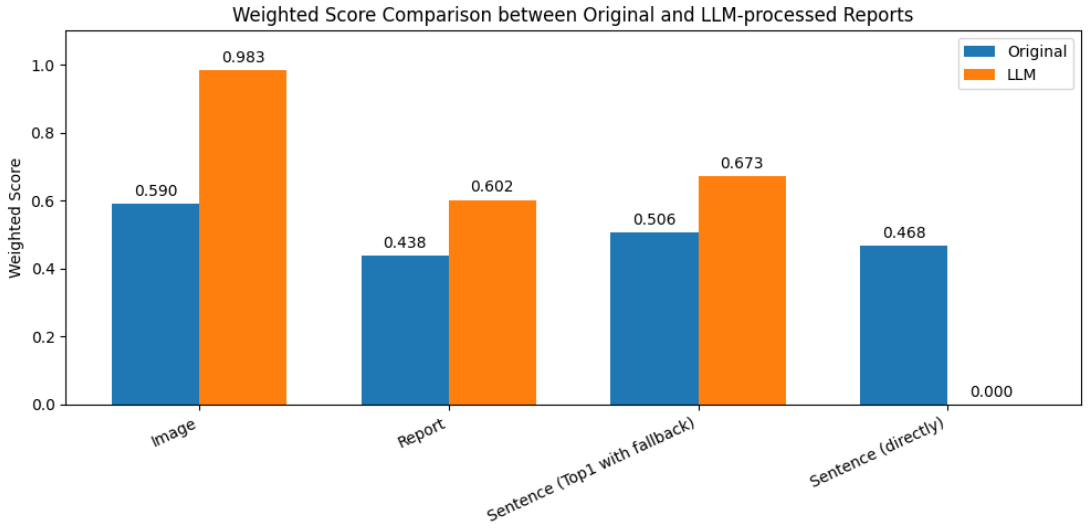
- **Compare LLM-processed reports with original biomedCLIP reports**
- **F1-macro score outperformed**

Experiment result (original and LLM-processed)

Experiment result(weighted score)

- **Combined F1 score and S_emb as weighted score:**
- *weigthed score = 0.65 · F1_norm + 0.35·S_norm*

- **LLM-processed reports outperform original biomedCLIP reports across all levels on weighted score**



Conclusion

Conducted experiments showing that BioMedCLIP outperforms the CLIP model.

Selected BioMedCLIP based on its superior performance.

Successfully built a dual-layer report generation system using BioMedCLIP and an LLM.

Generated accurate and fluent X-ray reports with our final system.

Future work

- Explore additional report generation methods to identify more effective and accurate approaches.
- Perform comprehensive evaluations to ensure the accuracy and reliability of the results.
- Conduct more detailed testing to improve system robustness and prevent failures in real-world use.
- Enhance the frontend with more user-friendly interaction features.

Reference

Endo, M., Krishnan, R., Krishna, V., Ng, A. Y. & Rajpurkar, P. (2021), Retrieval-based chest x-ray report generation using a pre-trained contrastive language-image model, *in* 'Proceedings of Machine Learning for Health (ML4H)', Vol. 158, PMLR, pp. 209–219.
URL: *https://proceedings.mlr.press/v158/endo21a.html*